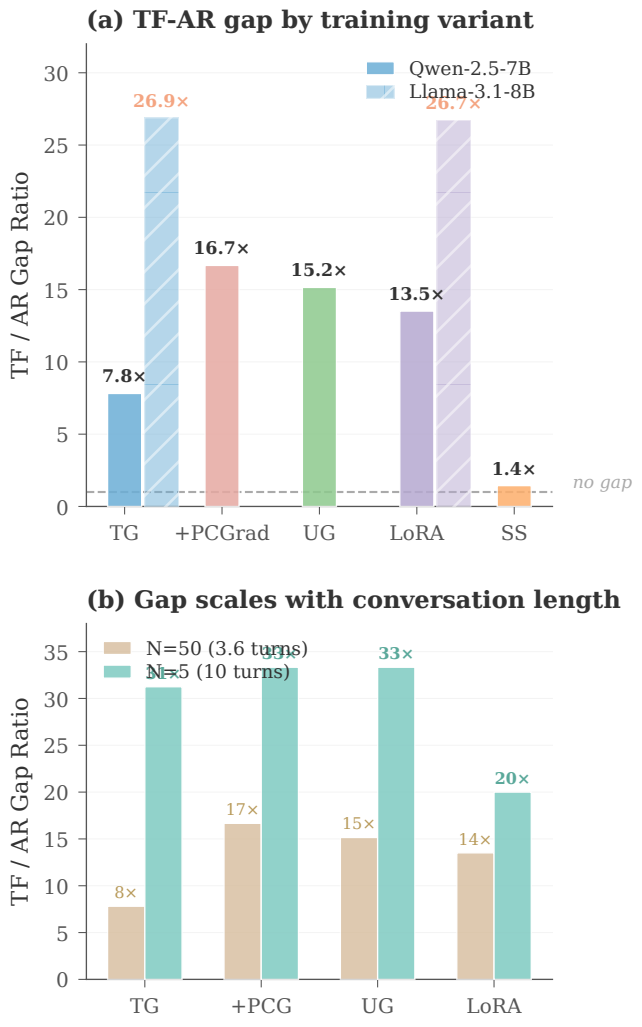
(a) TF-AR gap by training variant
Qwen-2.5-7B
Llama-3.1-8B
TF / AR Gap Ratio
26.9×
26.7×
16.7×
15.2×
13.5×
7.8×
1.4×
no gap
TG
+PCGrad
UG
LoRA
SS
(b) Gap scales with conversation length
N=50 (3.6 turns)
N=5 (10 turns)
TF / AR Gap Ratio
8×
17×
15×
14×
33×
20×
TG
+PCG
UG
LoRA